Prompt for Control Flow Graph Construction

You are a static program analysis engine tasked with constructing a Control Flow Graph (CFG) for the given Python source code.

Requirements

1. Nodes: Each node represents at most one statement or a expression-level execution unit, with a single entry point, and a single exit point.

2. Edges: The directed edges represent possible control transfers between nodes. Explicitly label edges with conditions (e.g., if condition=True/False), and include fall-through edges, loop back-edges, exceptional edges.

3. Language Constructs: The CFG construction should handle Python language constructs, including but not limited to conditionals, loops, function-level control flow, exception handling, and context managers.

4. Entry and Exit Nodes. Create a single Entry node representing function start, and a single **Exit** node representing normal termination. All return statements must connect to the Exit node.

Now you will be given input as a function, output the CFG in the following format: